

Avighna Chhatrapati

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EDUCATION

University of Illinois Urbana-Champaign

Aug 2026 – May 2030

Bachelor of Science in Computer Science

Coursework: Algorithms & Models of Computation, Data Structures, Discrete Structures, Multivariable Calculus

AWARDS

Asia-Pacific Informatics Olympiad (Taiwan) (May 2026) – Bronze medal

Indian National Olympiad in Informatics – Two silver medals in 2024 and 2025, and a gold medal in 2026 (top 30 and top 10 nationally respectively)

Indian National Mathematical Olympiad – Qualified for the exam (top 0.5% out of ~100,000 participants)

EXPERIENCE

Shortest Path to IOI

Jul 2024 – Present

Founder and Lead Trainer

- Founded and currently lead a free mentorship program for students preparing for the International Olympiad in Informatics (IOI), endorsed by IARCS (India's official informatics olympiad agency)
- Secured \$11,500 in funding to run and grow the program
- Design and deliver lectures, set mock contests, and train students

PROJECTS

libapfs

C++, Systems Programming, Filesystems

- An open-source, read-only implementation of the Apple File System (APFS) for Windows, Linux, and macOS, supporting browsing, extraction, and volume mounting
- Implemented parsing of APFS containers, volumes, B-trees, object maps, directory entries, and file extents from raw disks and disk images
- Added native filesystem mounting through WinFsp on Windows and FUSE on Linux, alongside a cross-platform CLI

oi-checklist

Node.js, TypeScript, Prisma, Python

- A full-stack web app to track progress across major Olympiads in Informatics, covering ~2,900 problems from 26 sources
- Used by 1,000+ users and recommended by USACO Guide as a competitive programming resource
- Implemented auto-sync with oj.uz, qoj.ac, and CodeChef, and virtual contests with live timers, medal cutoffs, and olympiad subtask scoring

arithmetica & basic-math-operations

x86-64 Assembly, C, C++, Python

- arithmetica is a symbolic algebra engine using an abstract syntax tree (AST) to represent and manipulate expressions, with support for simplification, differentiation, and equation solving
- basic-math-operations is an arbitrary-precision arithmetic library supporting integer and rational arithmetic with configurable precision
- Implemented core arithmetic algorithms in x86-64 assembly for Windows and Linux and exposed the library to C, C++, and Python

SKILLS

Languages: C, C++, x86-64 Assembly, Java, TypeScript, Python, SQL